

ME-203 THERMAL ENGINEERING-II

L	T	P	Credit	Area		CWS	PRS	MTE	ETE	PRE
3	0	2	4	DCC		15	25	20	40	-

Course objectives:

The primary objective of this course is to provide students with a comprehensive theoretical and practical understanding of thermodynamic cycles and principles governing the operation, performance, and design of various energy conversion devices, including reciprocating and centrifugal compressors, internal combustion (IC) engines, gas turbines, and jet propulsion systems.

Course Outcomes: After completing the course, student will be able to

CO1	Understand the working principle of reciprocating air compressor and multistage compression.
CO2	Understand the working principle of centrifugal compressor and compressor performance and vacuum pump.
CO3	Analyze the various air standard cycles, performance, comparison and its application.
CO4	Know the basic working principle of Internal combustion engines and analyze the combustion theory.
CO5	Describe the working principle and analysis of gas turbine cycles with intercooling, reheating and regeneration including performance and application of gas turbines.
CO6	Describe the theory of Jet Propulsion and its various application, theory of Rocket engines and its different kinds.

Syllabus		Contact Hours
Unit-1	Reciprocating Air Compressor: Steady flow analysis, isothermal, adiabatic and polytropic compression; single-and multi-stage compression, ideal intermediate pressure; compressor clearance, volumetric and isothermal efficiency; minimum work requirement of a compressor.	6
Unit-2	Centrifugal compressor: Velocity diagrams, efficiency of compressor stage, choice of reaction stage pressure rise, surging, multi-stage compressor, compressor performance, vacuum pump.	5
Unit-3	Gas Power Cycles: Air standard cycle, Otto, Diesel and dual cycles, P-V and T-s diagrams of these cycles, efficiency, mean effective pressure. Comparison of Otto, Diesel, dual cycles for same compression ratio and heat input, Stirling cycles, Ericsson cycle, Atkinson cycle, basic gas turbine cycle (Brayton cycle) for open and closed systems, efficiency of gas turbine cycle.	7
Unit-4	Introduction to I.C Engines: Classification; two and four stroke, SI and CI engine its parts, working principle and valve and port timing diagram, Principles of combustion, knocking and emissions in SI engine, theory of detonation, Principles of combustion, knocking and emissions in CI engine.	9
Unit-5	Gas Turbines: Simple open and closed cycle gas turbine, efficiency and specific output of simple cycle, effects of regeneration, re-heating and intercooling on efficiency and work output, effect of operating variables on thermal efficiency, air rate, work ratio; water injection, Advantages and disadvantages of gas turbine; gas turbine components, performance and application of gas turbine.	9

Unit-6	Jet Propulsion: Introduction to jet propulsion, advantages and disadvantages of jet propulsion, turbojet engine with and without after burner, turboprop, ramjet, pulsejet, rocket engines–operation, solid and Liquid propellants.	6
	Total	42

Reference Book:	
1	Basic and Applied Thermodynamics by P K Nag, Tata McGraw Hill Education private limited
2	Thermodynamics: An Engineering Approach by Yunus A Cengel and Michael A Boles, MacGraw Hill Education
3	Thermal Engineering by Mahesh M Rathore, Tata McGraw Hill Education private limited.
4	Thermal Engineering by P L Ballney, Khanna Publishers
5	Engineering Thermodynamics by D S Kumar S K Kataria and Sons
6	Power Plant Engineering by P K Nag, Tata McGraw Hill Education private limited

CO-PO/PSO Matrix

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3
CO1	3	2	2	1	1	1	2	1	1	1	1	2	3	2	1
CO2	3	2	2	1	2	2	2	1	1	2	2	2	3	2	1
CO3	3	3	3	2	1	1	2	1	2	2	1	2	2	3	2
CO4	3	3	3	2	1	1	2	1	1	1	1	2	3	2	1
CO5	3	3	2	2	1	1	2	1	1	1	1	2	3	2	1
CO6	3	2	2	2	1	1	2	1	2	1	1	2	2	2	1